Strong Law of Large Numbers

Stats 220A

What the SLLN adds

WLLN: $S_n/n \xrightarrow{\mathbb{P}} \mu$. Equivalently, for every $\varepsilon > 0$,

$$\mathbb{P}\left(\left|\frac{S_n}{n} - \mu\right| > \varepsilon\right) \to 0.$$

This controls the *probability* of deviation at each n, but does not assert that a given sample path eventually stays close to μ.

SLLN: $S_n/n \xrightarrow{\text{a.s.}} \mu$. For *almost every* ω, the sequence $S_n(\omega)/n$ converges to μ.

The difference. WLLN allows the bad set to drift with n — different ω's can fail at different times. SLLN requires a single null set that covers all failures simultaneously.

The surprise. Etemadi (1981) proved the SLLN under *pairwise* independence. Full mutual independence is not needed.

Compare: the WLLN often follows under uncorrelatedness (via an L^2 argument). Etemadi's SLLN needs pairwise independence but not full independence.

Etemadi's SLLN: statement

Theorem (Etemadi 1981)

Let $X_1, X_2, \ldots$ be **pairwise independent** *and identically distributed with $\mathbb{E}|X_1| < \infty$. Let $\mu = \mathbb{E}X_1$ and $S_n = X_1 + \cdots + X_n$. Then*

$$\frac{S_n}{n} \xrightarrow{\text{a.s.}} \mu.$$

Proof strategy. Three steps:

1. **Truncation.** Replace X_k by $Y_k = X_k \mathbf{1}_{\{|X_k| \le k\}}$. Show S_n and $\tilde{S}_n = \sum_k Y_k$ agree eventually a.s.
2. **Subsequence.** Prove $\tilde{S}_{n_k}/n_k \xrightarrow{\text{a.s.}} \mu$ along $n_k = \lceil \alpha^k \rceil$ for $\alpha > 1$ using a variance bound and Borel–Cantelli.
3. **Interpolation.** Fill in between subsequence values using monotonicity of partial sums (after passing to nonneg parts).

Each step is elementary; the cleverness is in choosing the right subsequence.

Step 1: truncation

Define $Y_k = X_k \mathbf{1}_{\{|X_k| \le k\}}$. The key fact:

Lemma

$\sum_{k=1}^{\infty} \mathbb{P}(X_k \neq Y_k) < \infty$.

Proof.

Since the X_k are i.i.d.,

$$\sum_{k=1}^{\infty} \mathbb{P}(|X_k| > k) = \sum_{k=1}^{\infty} \mathbb{P}(|X_1| > k).$$

Moreover,

$$\sum_{k=1}^{\infty} \mathbb{P}(|X_1| > k) \le \sum_{k=1}^{\infty} \int_{k-1}^{k} \mathbb{P}(|X_1| > t)\, dt = \int_0^{\infty} \mathbb{P}(|X_1| > t)\, dt = \mathbb{E}|X_1| < \infty.$$

Hence $\sum_k \mathbb{P}(X_k \neq Y_k) < \infty$, so by BC1, $\mathbb{P}(X_k \neq Y_k \text{ i.o.}) = 0$. Therefore there exists $K(\omega) < \infty$ such that for all $k \ge K(\omega)$, $X_k(\omega) = Y_k(\omega)$, and thus $\tilde{S}_n = \sum_{k=1}^{n} Y_k$ eventually agree a.s., and it suffices to show $\tilde{S}_n/n \xrightarrow{\text{a.s.}} \mu$. □

Note. This is the same truncation idea as in the WLLN proof, now applied to get an a.s. result via BC1 rather than a probabilistic bound.

A key variance summability bound (Durrett/Etemadi)

Lemma

Let $Y_k = X_k \mathbf{1}_{\{|X_k| \le k\}}$. *Then*

$$\sum_{k=1}^{\infty} \frac{\mathrm{Var}(Y_k)}{k^2} \;\le\; 4\,\mathbb{E}|X_1|.$$

Proof.

Since $\mathrm{Var}(Y_k) \le \mathbb{E}[Y_k^2]$ and $|Y_k| \le k$,

$$\mathbb{E}[Y_k^2] = \int_0^{\infty} 2y\,\mathbb{P}(|Y_k| > y)\,dy \le \int_0^{k} 2y\,\mathbb{P}(|X_1| > y)\,dy.$$

Therefore, by Tonelli/Fubini (nonnegative terms),

$$\sum_{k=1}^{\infty} \frac{\mathbb{E}[Y_k^2]}{k^2} \le \int_0^{\infty} \Big(\sum_{k>y} k^{-2}\Big)\, 2y\,\mathbb{P}(|X_1| > y)\,dy.$$

For all $y \ge 0$, one checks the crude bound $2y \sum_{k>y} k^{-2} \le 4$. Hence

$$\sum_{k=1}^{\infty} \frac{\mathrm{Var}(Y_k)}{k^2} \le \sum_{k=1}^{\infty} \frac{\mathbb{E}[Y_k^2]}{k^2} \le 4 \int_0^{\infty} \mathbb{P}(|X_1| > y)\,dy = 4\mathbb{E}|X_1|.$$

Step 2: a geometric subsequence + Borel–Cantelli

Fix $\alpha > 1$ and define $k(n) = \lfloor \alpha^n \rfloor$. Let $\tilde{S}_m = \sum_{j=1}^{m} Y_j$ and $\tilde{\mu}_m = \mathbb{E}[\tilde{S}_m]$.

Lemma

For every $\varepsilon > 0$,

$$\sum_{n=1}^{\infty} \mathbb{P}\left(\left| \tilde{S}_{k(n)} - \tilde{\mu}_{k(n)} \right| > \varepsilon k(n) \right) < \infty,$$

hence

$$\frac{\tilde{S}_{k(n)} - \tilde{\mu}_{k(n)}}{k(n)} \xrightarrow{\text{a.s.}} 0.$$

Conclusion. Since $\mathbb{E} Y_m \to \mu$ and Cesàro averaging preserves limits,

$$\frac{\tilde{\mu}_{k(n)}}{k(n)} = \frac{1}{k(n)} \sum_{m=1}^{k(n)} \mathbb{E} Y_m \to \mu,$$

so $\tilde{S}_{k(n)}/k(n) \xrightarrow{\text{a.s.}} \mu$.

Proof of Lemma

Proof.

By Chebyshev and pairwise independence (variance additivity),

$$\mathbb{P}\Big(\Big|\tilde{S}_{k(n)} - \tilde{\mu}_{k(n)}\Big| > \varepsilon k(n)\Big) \le \frac{\mathrm{Var}(\tilde{S}_{k(n)})}{\varepsilon^2 k(n)^2} = \frac{1}{\varepsilon^2}\sum_{m=1}^{k(n)} \frac{\mathrm{Var}(Y_m)}{k(n)^2}.$$

Sum over n and swap sums (Tonelli):

$$\sum_{n=1}^{\infty} \mathbb{P}(\cdots) \le \frac{1}{\varepsilon^2}\sum_{m=1}^{\infty} \mathrm{Var}(Y_m) \sum_{n:\, k(n)\ge m} \frac{1}{k(n)^2}.$$

Since $k(n) \asymp \alpha^n$, the inner tail is a geometric series and satisfies

$$\sum_{n:\, k(n)\ge m} \frac{1}{k(n)^2} \lesssim \frac{C_\alpha}{m^2}.$$

Thus

$$\sum_{n=1}^{\infty} \mathbb{P}(\cdots) \lesssim \frac{C_\alpha}{\varepsilon^2}\sum_{m=1}^{\infty} \frac{\mathrm{Var}(Y_m)}{m^2} < \infty$$

by the previous lemma. By BC1, we have the proof.

Step 3: interpolation

Reduce to the case $X_k \geq 0$ by writing $X_k = X_k^+ - X_k^-$ and applying the argument to (X_k^+) and (X_k^-) separately.

Fix $\alpha > 1$ and set $n_j = \lceil \alpha^j \rceil$ (equivalently, $n_j = k(j)$ up to rounding). From Step 2 we have $\tilde{S}_{n_j}/n_j \xrightarrow{\text{a.s.}} \mu$.
For $n_j \leq n < n_{j+1}$, since $\tilde{S}_n$ is nondecreasing,

$$\frac{\tilde{S}_{n_j}}{n_{j+1}} \leq \frac{\tilde{S}_n}{n} \leq \frac{\tilde{S}_{n_{j+1}}}{n_j}.$$

Since $n_{j+1}/n_j \to \alpha$, we obtain

$$\frac{1}{\alpha}\mu \leq \liminf_{n\to\infty} \frac{\tilde{S}_n}{n} \leq \limsup_{n\to\infty} \frac{\tilde{S}_n}{n} \leq \alpha\mu \quad \text{a.s.}$$

Letting $\alpha \downarrow 1$ along a countable sequence (so the exceptional null set stays null) yields $\tilde{S}_n/n \xrightarrow{\text{a.s.}} \mu$.

Why a geometric subsequence? It is sparse enough that the Borel–Cantelli sum in Step 2 is finite (via a geometric-series tail bound), but dense enough that consecutive terms satisfy $n_{j+1}/n_j \approx \alpha$, which makes the interpolation work.

Necessity of $\mathbb{E}|X_1| < \infty$

Etemadi's theorem requires $\mathbb{E}|X_1| < \infty$. The condition is sharp.

Theorem (Durrett 2.3.8)

Let $X_1, X_2, \ldots$ be i.i.d. with $\mathbb{E}|X_1| = \infty$. Then

$$\mathbb{P}(|X_n| \geq n \text{ i.o.}) = 1,$$

and consequently $\mathbb{P}(\lim_{n\to\infty} S_n/n$ exists in $(-\infty, \infty)) = 0$.

Proof.

From the layer-cake formula, $\mathbb{E}|X_1| = \int_0^\infty \mathbb{P}(|X_1| > t)\, dt \leq \sum_{n=0}^\infty \mathbb{P}(|X_1| > n)$. Since $\mathbb{E}|X_1| = \infty$, we get $\sum_n \mathbb{P}(|X_n| > n) = \infty$. By BC2 (events are independent), $\mathbb{P}(|X_n| \geq n \text{ i.o.}) = 1$.

For the second claim: on $C = \{\lim S_n/n \text{ exists finitely}\}$, we have $S_n/(n(n+1)) \to 0$, so

$$\frac{S_{n+1}}{n+1} - \frac{S_n}{n} = \frac{X_{n+1}}{n+1} - \frac{S_n}{n(n+1)} \to 0.$$

But on $C \cap \{|X_n| \geq n \text{ i.o.}\}$, $|X_{n+1}/(n+1)| \geq 1/(1+1/n)$ i.o., a contradiction. Hence $C \cap \{|X_n| \geq n \text{ i.o.}\} = \emptyset$, and since the latter has probability 1, $\mathbb{P}(C) = 0$. $\square$

Why we need a maximal inequality

In the SLLN proof above, we controlled S_n at a fixed n (or along a subsequence). But many arguments require controlling $\max_{k \le n} |S_k|$ simultaneously.

The naive bound is $\mathbb{P}(\max_{k \le n} |S_k| > x) \le \sum_{k=1}^{n} \mathbb{P}(|S_k| > x)$, but this loses a factor of n.

Kolmogorov's idea. For independent summands, we can do much better: the maximum partial sum satisfies the *same* type of bound as the terminal sum, up to a constant.

Theorem (Kolmogorov's Maximal Inequality)

Let $X_1, \ldots, X_n$ be independent with $\mathbb{E}X_k = 0$ and $\mathbb{E}X_k^2 < \infty$. Then for any $x > 0$:

$$\mathbb{P}\left(\max_{1 \le k \le n} |S_k| \ge x\right) \le \frac{\mathrm{Var}(S_n)}{x^2}.$$

The right side depends only on $\mathrm{Var}(S_n)$, not on $n \cdot \max_k \mathrm{Var}(S_k)$.

Proof of Kolmogorov's maximal inequality

Let $A = \{\max_{k \le n} |S_k| \ge x\}$ and $A_k = \{|S_1| < x, \dots, |S_{k-1}| < x, |S_k| \ge x\}$, the event that k is the *first* time $|S_k| \ge x$. The A_k are disjoint and $A = \bigcup_k A_k$.
On A_k, we have $|S_k| \ge x$. Write $S_n = S_k + (S_n - S_k)$, so $S_n^2 \ge S_k^2 + 2S_k(S_n - S_k)$. Integrating over A_k:

$$\int_{A_k} S_n^2 \, d\mathbb{P} \ge \int_{A_k} S_k^2 \, d\mathbb{P} + 2 \int_{A_k} S_k(S_n - S_k) \, d\mathbb{P}.$$

The second term vanishes: A_k depends only on $X_1, \dots, X_k$, and $S_n - S_k = X_{k+1} + \cdots + X_n$ is independent of $\mathcal{F}_k$ with mean zero, so $\mathbb{E}[\mathbf{1}_{A_k} S_k(S_n - S_k)] = \mathbb{E}[\mathbf{1}_{A_k} S_k] \cdot \mathbb{E}[S_n - S_k] = 0$.
Therefore $\int_{A_k} S_n^2 \, d\mathbb{P} \ge x^2 \mathbb{P}(A_k)$. Summing over k:

$$\mathrm{Var}(S_n) = \mathbb{E}[S_n^2] \ge \sum_{k=1}^{n} \int_{A_k} S_n^2 \, d\mathbb{P} \ge x^2 \mathbb{P}(A). \quad \square$$

Application 1: Glivenko–Cantelli theorem

Definition

The **empirical CDF** of $X_1, \ldots, X_n$ i.i.d. with CDF F is $F_n(x) = \frac{1}{n}\sum_{k=1}^{n} \mathbf{1}_{\{X_k \le x\}}$.

Theorem (Glivenko–Cantelli)

$\sup_{x\in\mathbb{R}} |F_n(x) - F(x)| \xrightarrow{\text{a.s.}} 0.$

Proof sketch.

For each fixed x, $F_n(x)$ is an average of i.i.d. Bernoulli$(F(x))$ variables, so $F_n(x) \xrightarrow{\text{a.s.}} F(x)$ by the SLLN. The hard part is uniformity over x.

Fix $\varepsilon > 0$ and choose a finite grid $-\infty = x_0 < x_1 < \cdots < x_m = \infty$ so that $F(x_j) - F(x_{j-1}^-) \le \varepsilon$ for each j. (Possible since F is right-continuous with left limits and has at most countably many jumps.) The SLLN gives $F_n(x_j) \xrightarrow{\text{a.s.}} F(x_j)$ for each j. For $x \in (x_{j-1}, x_j)$, monotonicity of F_n and F gives

$$F_n(x) \le F_n(x_j^-) \le F(x_j^-) + \varepsilon \le F(x) + 2\varepsilon$$

and $F_n(x) \ge F(x) - 2\varepsilon$ by an analogous lower bound. Hence $\sup_x |F_n(x) - F(x)| \le 2\varepsilon$ eventually a.s. $\square$

Statistical meaning. The empirical CDF is a *uniformly consistent* estimator of F. The entire distribution is recoverable from data.

Application 2: renewal theory

Let $X_1, X_2, \ldots$ be i.i.d. with $0 < X_i < \infty$ a.s. and $\mathbb{E}X_1 = \mu \leq \infty$. Set $T_n = X_1 + \cdots + X_n$ (the time of the nth renewal) and

$$N_t = \sup\{n \geq 0 : T_n \leq t\}, \quad t \geq 0,$$

the number of renewals in $[0, t]$.

Theorem (Elementary renewal theorem, a.s. version)

$N_t/t \xrightarrow{\text{a.s.}} 1/\mu$ as $t \to \infty$ (with $1/\infty = 0$).

Proof.

By the SLLN, $T_n/n \xrightarrow{\text{a.s.}} \mu$. By definition of N_t, $T_{N_t} \leq t < T_{N_t+1}$, so dividing by N_t:

$$\frac{T_{N_t}}{N_t} \leq \frac{t}{N_t} \leq \frac{T_{N_t+1}}{N_t+1} \cdot \frac{N_t+1}{N_t}.$$

Since $T_n < \infty$ for all n, we have $N_t \uparrow \infty$ as $t \to \infty$. The SLLN implies $T_{N_t}/N_t \xrightarrow{\text{a.s.}} \mu$ (a.s. convergence is preserved under random time-changes when $N_t \to \infty$ a.s.), and $(N_t + 1)/N_t \to 1$. Squeezing gives $t/N_t \xrightarrow{\text{a.s.}} \mu$. □

Warning. The analogous statement fails for convergence in probability: if $X_n \xrightarrow{\mathbb{P}} 0$ and $N(n) = \inf\{m \geq n : X_m = 1\}$, then $X_{N(n)} = 1$ a.s.

Application 3: Shannon's theorem

Let $X_1, X_2, \ldots$ be i.i.d. taking values in $\{1, \ldots, r\}$ with $\mathbb{P}(X_i = k) = p(k) > 0$. Think of $1, \ldots, r$ as letters of an alphabet; $X_1, X_2, \ldots$ is the output of an i.i.d. source.
Define the **path probability**

$$\pi_n(\omega) = p(X_1(\omega)) \cdots p(X_n(\omega)).$$

Since $\log \pi_n = \sum_{i=1}^n \log p(X_i)$ is a sum of i.i.d. random variables with $\mathbb{E}[\log p(X_1)] = \sum_k p(k) \log p(k) > -\infty$, the SLLN gives immediately:

Theorem (Shannon–McMillan, i.i.d. case)

$$-\frac{1}{n} \log \pi_n(\omega) \xrightarrow{\text{a.s.}} H \quad \textit{where} \quad H = -\sum_{k=1}^{r} p(k) \log p(k).$$

H is the **entropy** of the source. The theorem says all observed sequences are eventually *typical*: their probability is $\approx e^{-nH}$.
Asymptotic equipartition property. For any $\varepsilon > 0$, as $n \to \infty$:

$$\mathbb{P}\Big(e^{-n(H+\varepsilon)} \le \pi_n(\omega) \le e^{-n(H-\varepsilon)}\Big) \to 1.$$

The source output concentrates on $\approx e^{nH}$ sequences, each of probability $\approx e^{-nH}$.

Theorem 2.3.9: a quantitative refinement of BC2

BC2 says: if $A_1, A_2, \ldots$ are independent with $\sum \mathbb{P}(A_n) = \infty$, then $\mathbb{P}(A_n \text{ i.o.}) = 1$. Theorem 2.3.9 sharpens this to an a.s. ratio limit, and weakens independence to pairwise.

Theorem (Durrett 2.3.9)

If $A_1, A_2, \ldots$ are **pairwise independent** *with $\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \infty$, then*

$$\frac{\sum_{m=1}^{n} \mathbf{1}_{A_m}}{\sum_{m=1}^{n} \mathbb{P}(A_m)} \xrightarrow{\text{a.s.}} 1.$$

Proof sketch.

Let $S_n = \sum_{m=1}^{n} \mathbf{1}_{A_m}$ and $\mu_n = \mathbb{E}S_n = \sum_{m=1}^{n} \mathbb{P}(A_m) \to \infty$. Pairwise independence gives $\mathrm{Var}(S_n) \le \mu_n$ (since $\mathrm{Var}(\mathbf{1}_{A_m}) \le \mathbb{P}(A_m)$). Chebyshev gives $S_n/\mu_n \xrightarrow{\mathbb{P}} 1$. To upgrade to a.s., take the subsequence $n_k = \inf\{n : \mu_n \ge k^2\}$; then $\mu_{n_k} \asymp k^2$ and $\sum_k \mathrm{Var}(S_{n_k})/\mu_{n_k}^2 \lesssim \sum_k k^{-2} < \infty$. BC1 gives $S_{n_k}/\mu_{n_k} \xrightarrow{\text{a.s.}} 1$. Monotonicity of S_n and the slow growth $\mu_{n_{k+1}}/\mu_{n_k} \to 1$ fill in the gaps. $\square$

Application: record values

Let $X_1, X_2, \ldots$ be i.i.d. with continuous CDF F. Call X_k a **record** if $X_k > \max(X_1, \ldots, X_{k-1})$ (with X_1 always a record). Let $A_k = \{X_k \text{ is a record}\}$ and $R_n = \sum_{k=1}^{n} \mathbf{1}_{A_k}$.

Claim. $A_1, A_2, \ldots$ are independent with $\mathbb{P}(A_k) = 1/k$.
Why? Since F is continuous the X_i are a.s. distinct. Conditionally on any ordering of $X_1, \ldots, X_k$, each of the $k!$ rankings is equally likely by symmetry. The event $A_k = \{X_k \text{ has rank 1 among } X_1, \ldots, X_k\}$ has probability $1/k$. Independence follows by an induction on conditioning on ranks of later variables.

Since $\sum_{k=1}^{n} 1/k \sim \log n = \infty$, Theorem 2.3.9 applies:

Theorem

$R_n / \log n \xrightarrow{\text{a.s.}} 1$.

Interpretation. The number of records up to time n grows like $\log n$, independent of the distribution F (as long as it is continuous). Records become increasingly rare: the kth record occurs at time $\approx e^k$.

Note. The result is distribution-free — F cancels completely from the proof.

Application: longest head run

Let $X_1, X_2, \ldots$ be i.i.d. with $\mathbb{P}(X_k = 1) = \mathbb{P}(X_k = -1) = 1/2$. Let $L_n = \max_{1 \le m \le n} \ell_m$ where ℓ_m is the length of the run of $+1$'s ending at time m.

Theorem (Durrett 2.3.12)

$L_n / \log_2 n \xrightarrow{\text{a.s.}} 1$.

Proof.

Upper bound. A run of $+1$'s of length $\ge (1+\varepsilon)\log_2 n$ starting at position m has probability $2^{-(1+\varepsilon)\log_2 n} = n^{-(1+\varepsilon)}$. By a union bound over $m \le n$ and BC1, $\mathbb{P}(\ell_n \ge (1+\varepsilon)\log_2 n \text{ i.o.}) = 0$, so $\limsup L_n / \log_2 n \le 1$ a.s.

Lower bound. Partition $\{1, \ldots, n\}$ into disjoint blocks of length $b_n = \lfloor (1-\varepsilon)\log_2 n \rfloor + 1$. The number of blocks is $\ge n/\log_2 n$. Each block is an all-$+1$ run with probability $2^{-b_n} \ge n^{-(1-\varepsilon)}/2$. These block events are independent, so

$$\mathbb{P}(L_n \le (1-\varepsilon)\log_2 n) \le \left(1 - \frac{n^{-(1-\varepsilon)}}{2}\right)^{n/\log_2 n} \le \exp\left(-\frac{n^{\varepsilon}}{2\log_2 n}\right),$$

which is summable. BC1 gives $\liminf L_n / \log_2 n \ge 1$ a.s. $\square$

Summary

Result	**Key idea**
Etemadi's SLLN	Truncation + geometric subsequence + BC1 + interpolation
Necessity of $\mathbb{E}\lvert X\rvert < \infty$	BC2 + telescoping argument on $S_n/n - S_{n+1}/(n+1)$
Kolmogorov maximal ineq.	Control $\max \lvert S_k\rvert$ via independence of increments
Glivenko–Cantelli	SLLN pointwise + uniform control via finite grid
Renewal theory	SLLN + random time-change + squeeze argument
Shannon's theorem	SLLN applied to $\log p(X_i)$; entropy as the a.s. limit
Theorem 2.3.9	Pairwise BC2 + subsequence + monotone interpolation
Record values	$\mathbb{P}(A_k) = 1/k$ independent; apply Thm 2.3.9
Longest head run	BC1 upper bound + independent blocks lower bound